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COLLEGE 117
April 13, 2026
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Essentialism & Injustice



Recap of last Wednesday:



Technology

- A designed tool that permits an action (e.g., solving or creating problems, intervening in the environment)
- Assumptions, beliefs, and values inform tech development in ways that can create injustice

Participatory Justice

- The direct participation of those affected by a particular decision in the decision-making process.

Recap of last Wednesday:



Distributive justice

- The equitable, or fair, allocation of resources, opportunities (including decision-making opportunities), and benefits within a society or context

Inclusive design

- Designing for individuals who are traditionally excluded to build frameworks that work better for everyone

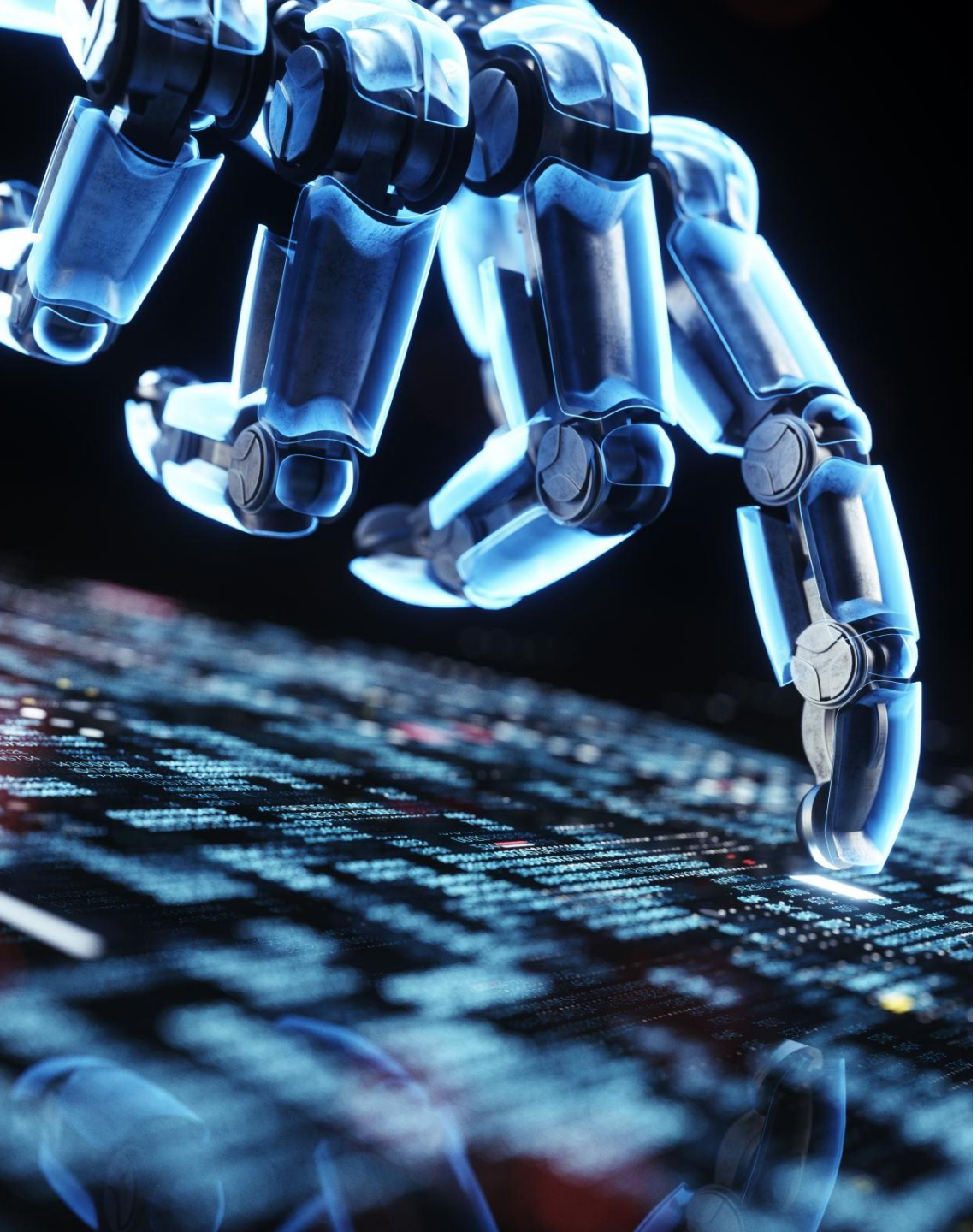


Questions from last week?

This week's goals:

Learn how societal and cultural phenomena shape health and well-being.

Understand how culturally constructed ideas– like "essentialism" can hinder justice.



Today's question:

How are science & technology used to create difference? What does this mean for justice?

Genetic determinism: the view that genes are fixed, immutable, and can be used to divide people into discrete, non-overlapping groups (Dar-Nimrod & Heine, 2011)

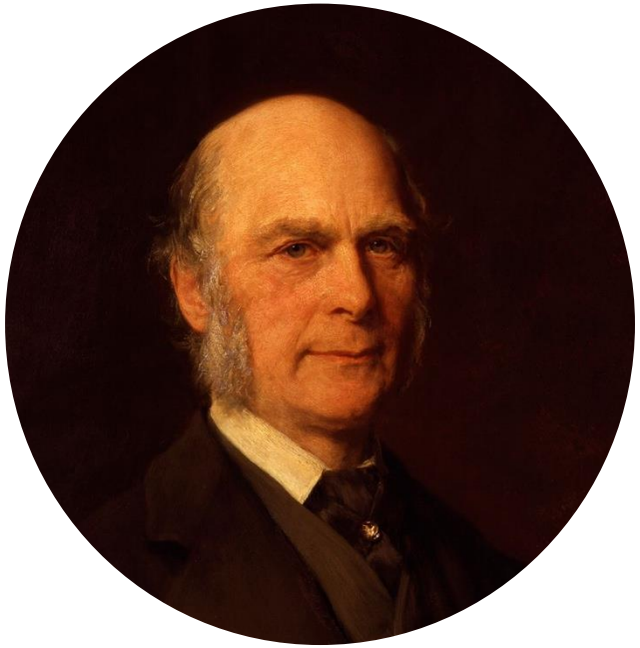




The ‘Gene’

“One of the most powerful and dangerous ideas in the history of science.” (Mukherjee, 2016)





Francis Galton

“I have no patience with the hypothesis occasionally expressed, and often implied, especially in tales written to teach children to be good, that babies are born pretty much alike, and that the sole agencies in creating differences between boy and boy, and man and man, are steady application and moral effort. It is in the most unqualified manner that I object to pretensions of natural equality.”

(1869)

The National Portrait Gallery, London,
taken from Britannica online



Arthur Estabrook Papers, Special Collections & Archives, University at Albany, SUNY taken from Encyclopedia Virginia

Carrie & Emma Buck

Buck v. Bell, 1927



Justice Oliver
Wendell Holmes Jr.

"It is better for all the world, if instead of waiting to execute degenerate offspring for crime, or to let them starve for their imbecility, society can prevent those who are manifestly unfit from continuing their kind."

Image: Britannica

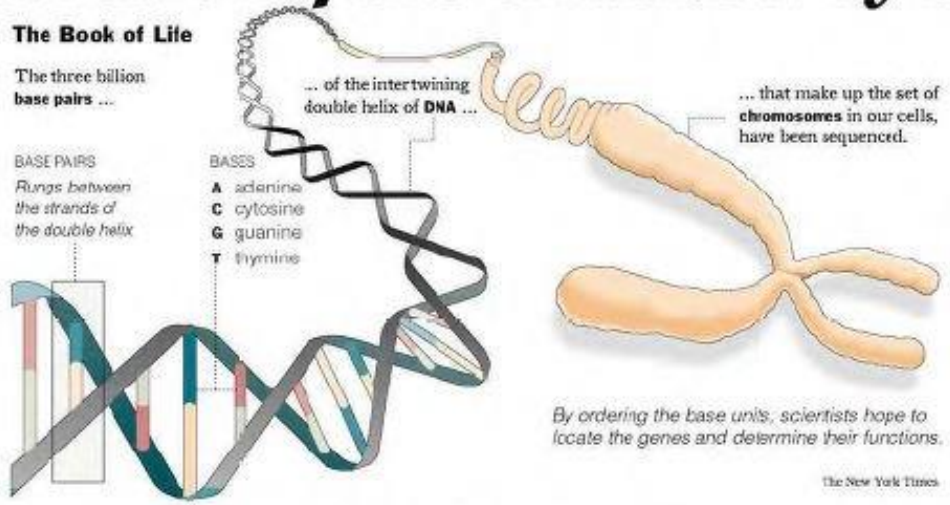
"All the News That's Fit to Print"

The New York Times

Late Edition
New York: **Today**, afternoon thunderstorms, high 88. **Tonight**, showers end, low 67. **Tomorrow**, partly cloudy with showers late, high 81. **Yesterday**, high 88, low 74. Weather map, Page D8.

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Genetic Code of Human Life Is Cracked by Scientists



Science Times
A special issue

- Putting the genome to work.
- Some information has already paid research dividends.
- Two research methods, two results.
- From Mendel to helix to genome.
- More articles, charts and photos of the genome effort.

Section F



Francis S. Collins, head of the Human Genome Project, left, with J. Craig Venter, head of Celera Genomics, after the announcement yesterday that they had finished the first survey of the human genome.

Paul Hosefros/The New York Times

A SHARED SUCCESS

2 Rivals' Announcement Marks New Medical Era, Risks and All

By NICHOLAS WADE

WASHINGTON, June 26 — In an achievement that represents a pinnacle of human self-knowledge, two rival groups of scientists said today that they had deciphered the hereditary script, the set of instructions that defines the human organism.

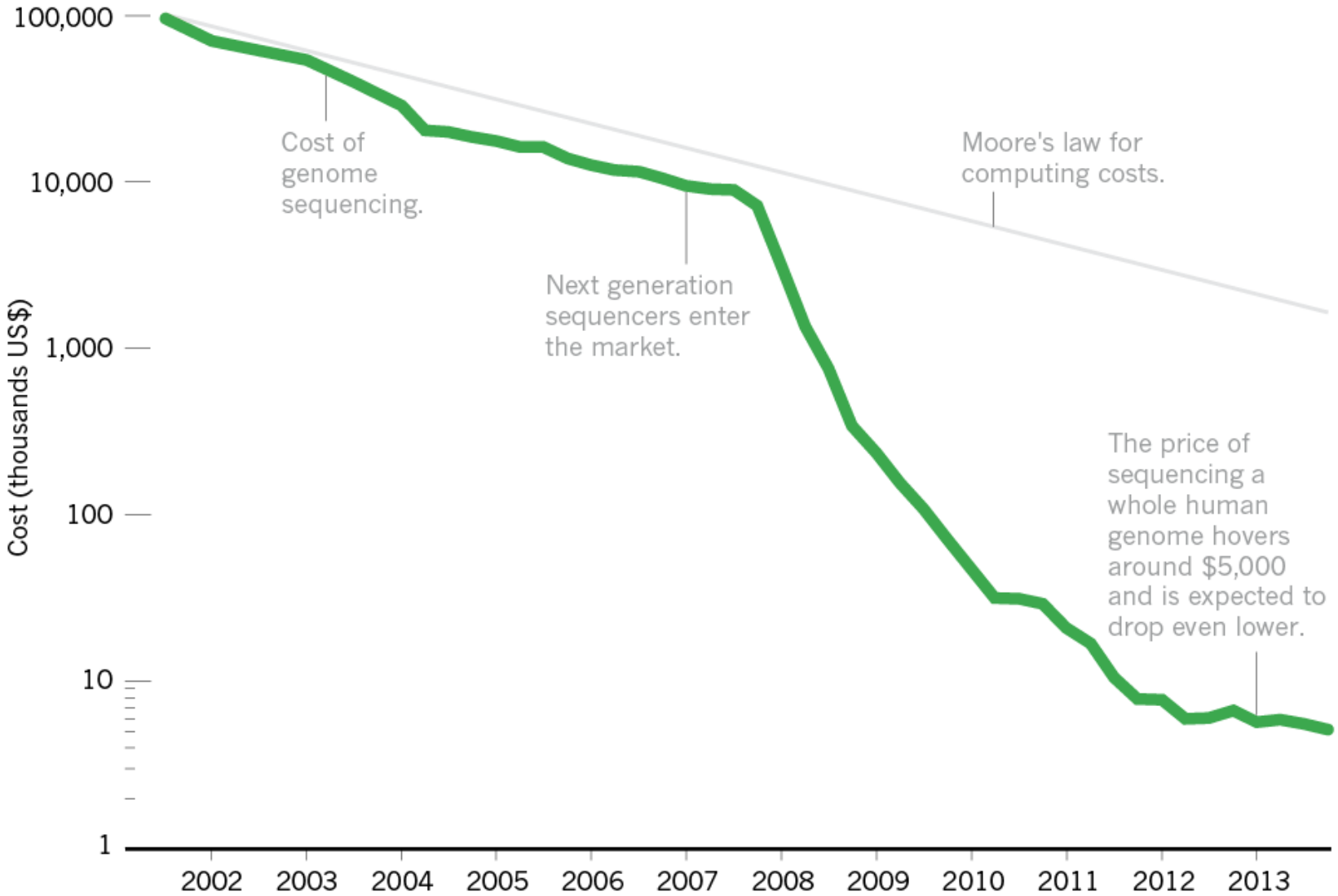
"Today we are learning the language in which God created life," President Clinton said at a White House ceremony attended by members of the two teams, Dr. James D. Watson, co-discoverer of the structure of DNA, and, via satellite, Prime Minister Tony Blair of Britain. [Excerpts, Page D8.]

The teams' leaders, Dr. J. Craig Venter, president of Celera Genomics, and Dr. Francis S. Collins, director of the National Human Genome Research Institute, praised each other's contributions and signaled a spirit of cooperation from now on, even though the two efforts will remain firmly independent.

The human genome, the ancient script that has now been deciphered, consists of two sets of 23 giant DNA

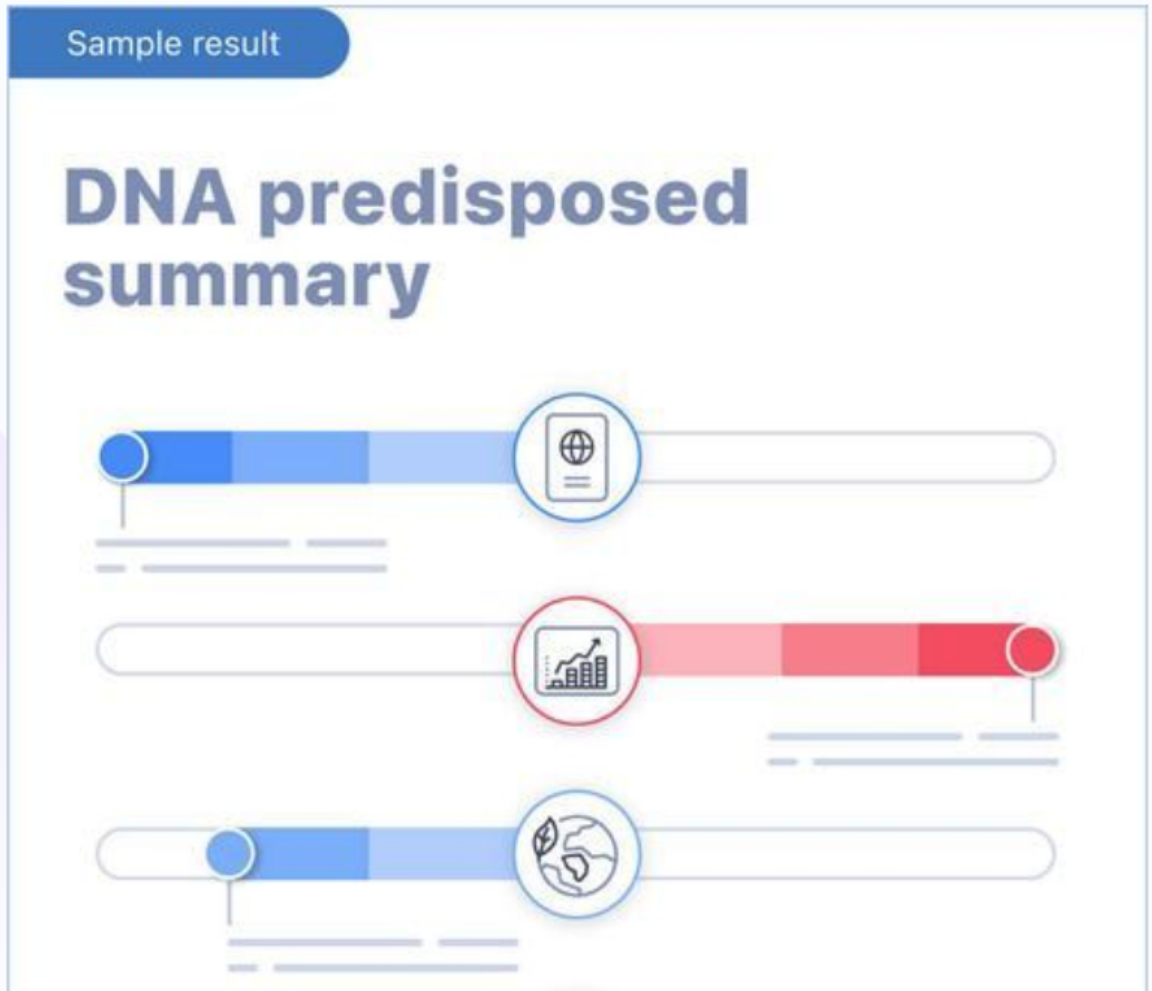
Falling fast

In the first few years after the end of the Human Genome Project, the cost of genome sequencing roughly followed Moore's law, which predicts exponential declines in computing costs. After 2007, sequencing costs dropped precipitously.



https://www.weforum.org/stories/2019/06/today-you-can-have-your-genome-sequenced-at-the-supermarket/

Are your political views affected by DNA?



However, it’s important to think about the practical potential of this kind of research. If our political views are affected by our DNA, we may be able to see across the political divide a little bit more clearly. Knowing that someone has a particular political perspective because of a characteristic determined by genetics can make it easier to understand their point of view.

Perhaps most importantly, knowing that a political affiliation DNA test can help determine someone's political viewpoints may make us stop trying to change each other's perspectives so aggressively. This could lead to much more civil conversations around politics.

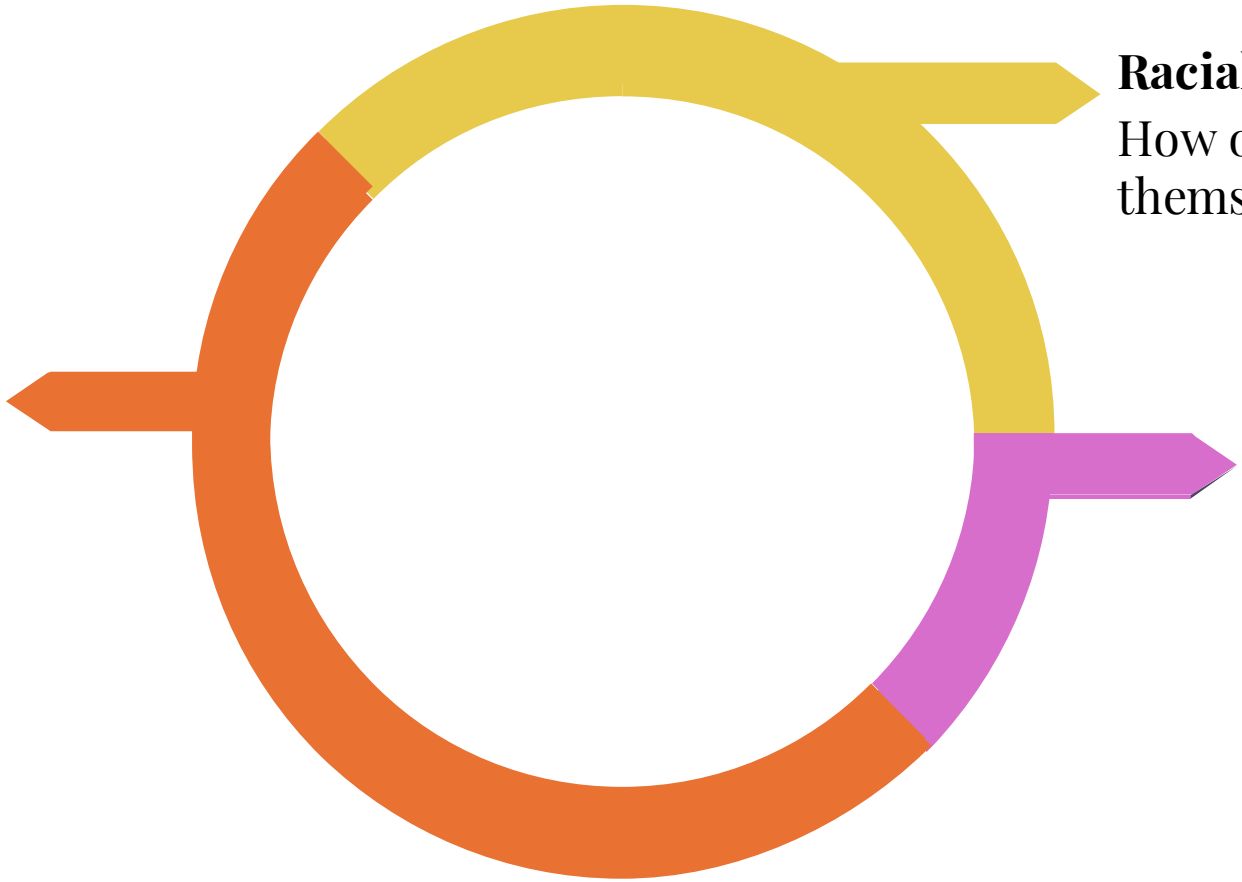
Racial essentialism: the view that members of a group share defining qualities, or essences, that are innate and unchanging and inform the nature, behavior, or abilities of group members (Roth, 2023)





Race

A fundamental organizing principle of many societies; a sociopolitical construct



Racial Identity
How one identifies themselves

Racial Classification
How society views, treats, understands, and engages with someone

Time & Context
How one identifies themselves and how one is identified by others is time and context dependent

1991	2020s
Brown	Mixed-Race (Pardo)
White	White (Branco)
Black	Black (Preto)
Indigenous	Indigenous (Indígina)
Yellow	Asian (Amarilo)

Brazilian census

1990

Aleut; Eskimo; Indian (Amer.)

Asian or Pacific Islander: Chinese; Filipino; Japanese; Korean; Vietnamese; Asian Indian; Other API

Black or Negro

Spanish/Hispanic Origin: Mexican, Mexican American, Chicano; Puerto Rican; Cuban; Other Spanish/Hispanic

Hawaiian; Guamanian; Samoan; Other API

White

Other Race

2020

White (+)

Black or African American (+)

American Indian or Alaska Native

Chinese

Japanese

Filipino

Korean

Asian Indian

Vietnamese

Other Asian

Native Hawaiian

Samoan

Chamorro

Other Pacific Islander

Mexican, Mexican American, Chicano (+)

Some other race

Puerto Rican

Cuban

Another Hispanic, Latino, Spanish Origin



Race is not biological, but what about traits like skin color that are biologically influenced?

A person's DNA influences a variety of aspects of their appearance. Some physical features, like skin color, are racialized: they carry social meaning that can tie a person to a racial classification.

In a different world, the DNA variants associated with skin pigmentation would not have the social power to cause differences in life outcomes.

It's the social structure and how it racializes phenotypes like skin color that *causes* differences between racial groups, not an unchanging aspect of biology.

Philadelphia, Yellow Fever Epidemic of 1793

- 1748, South Carolina physician John Lining: "There is something very singular in the constitution of the Negroes, which renders them not liable to this fever."

- "Over time, the belief that yellow fever was a disease to which Black people were immune, because of their race, took hold—and such a belief became fodder for slavery apologists." (Hogarth, 2019)



Image: Dialogue Institute



Absalom Jones & Richard Allen

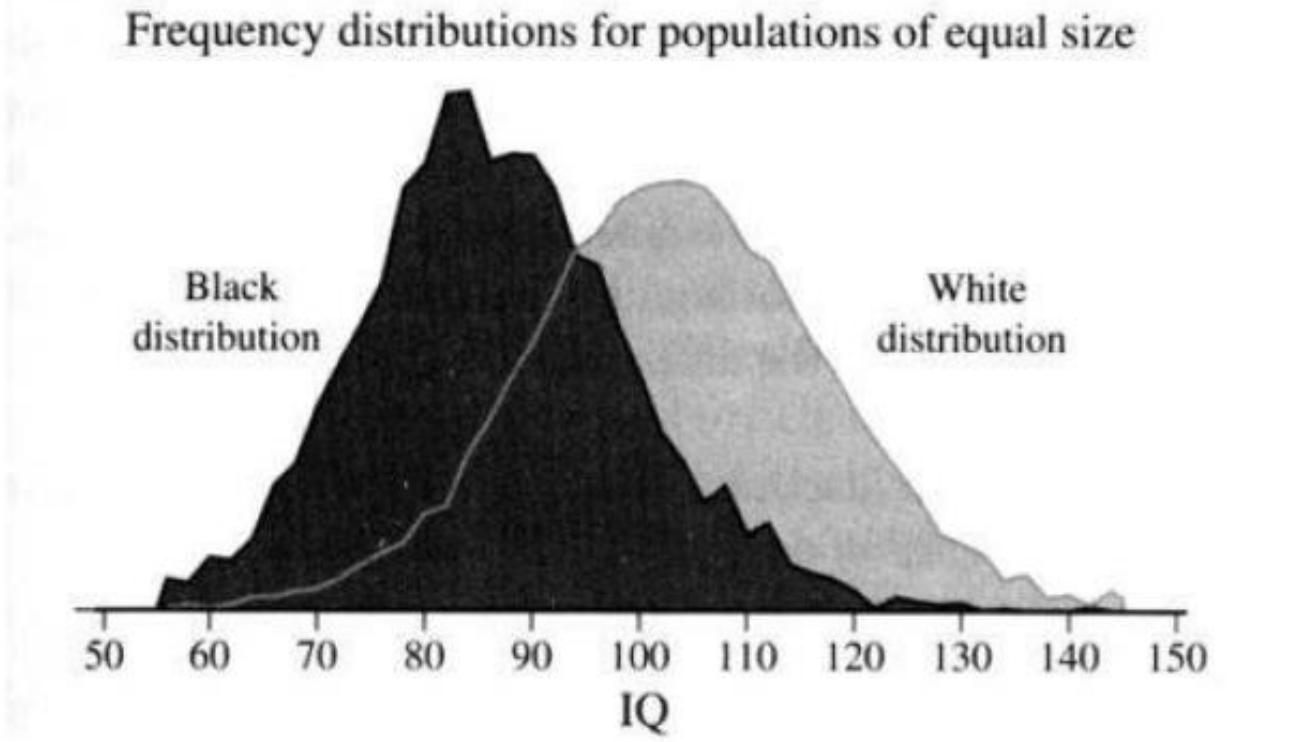
“The public were informed that in the West Indies and other places where this terrible malady had been, it was observed the blacks were not affected with it. Happy would it have been for you, and much more so for us, if this observation had been verified by our experience”

(1794)

Physical traits do not matter that much in the society of advanced human civilization. It has mattered in the past when humans had to plow the land and relied on their physical abilities to do work. But in this age of technology physical attributes matter less and less. I believe that the most important part of a human is the brain. The human brain is an amazing tool and is the reason why we as humans are predators of all and prey of none. It is obvious that people differ in cognitive abilities, this is undeniable and a proven fact. People with a "higher intelligence" often succeed in life by getting higher education and more professional careers, which often leads to greater contributions to human development in their lifetime. There is an undeniable difference between the brains of races and I will be explaining some of the differences below.

1: Intelligence and Genetics

The black and white IQ distributions in the NLSY, Version I



Strategic essentialism:
People draw upon multiple categories and causal factors in strategic ways based on the context. That is, given the multiple sets of information individuals carry around, genetics are used in ways that serve people's pre-existing beliefs and agendas (Condit, 2019)







Stephen Voss, taken from Politico

THE SOCIAL LIFE OF DNA

RACE, REPARATIONS, AND RECONCILIATION AFTER THE GENOME

Alondra Nelson

"Eye-opening, provocative, and deeply humane."
—Isabel Wilkerson

”For me, you know, **my brain is fully developed** at this point so, for me, my focus is my kids because **their brains are not fully developed** and so that’s why I find it interesting in general.”

– DTCP P01


Technology is a tool, or range of tools for doing something

Partner discussion:

How might essentialism create injustice? Can you think of any biotechnologies with essentialism incorporated into the design?





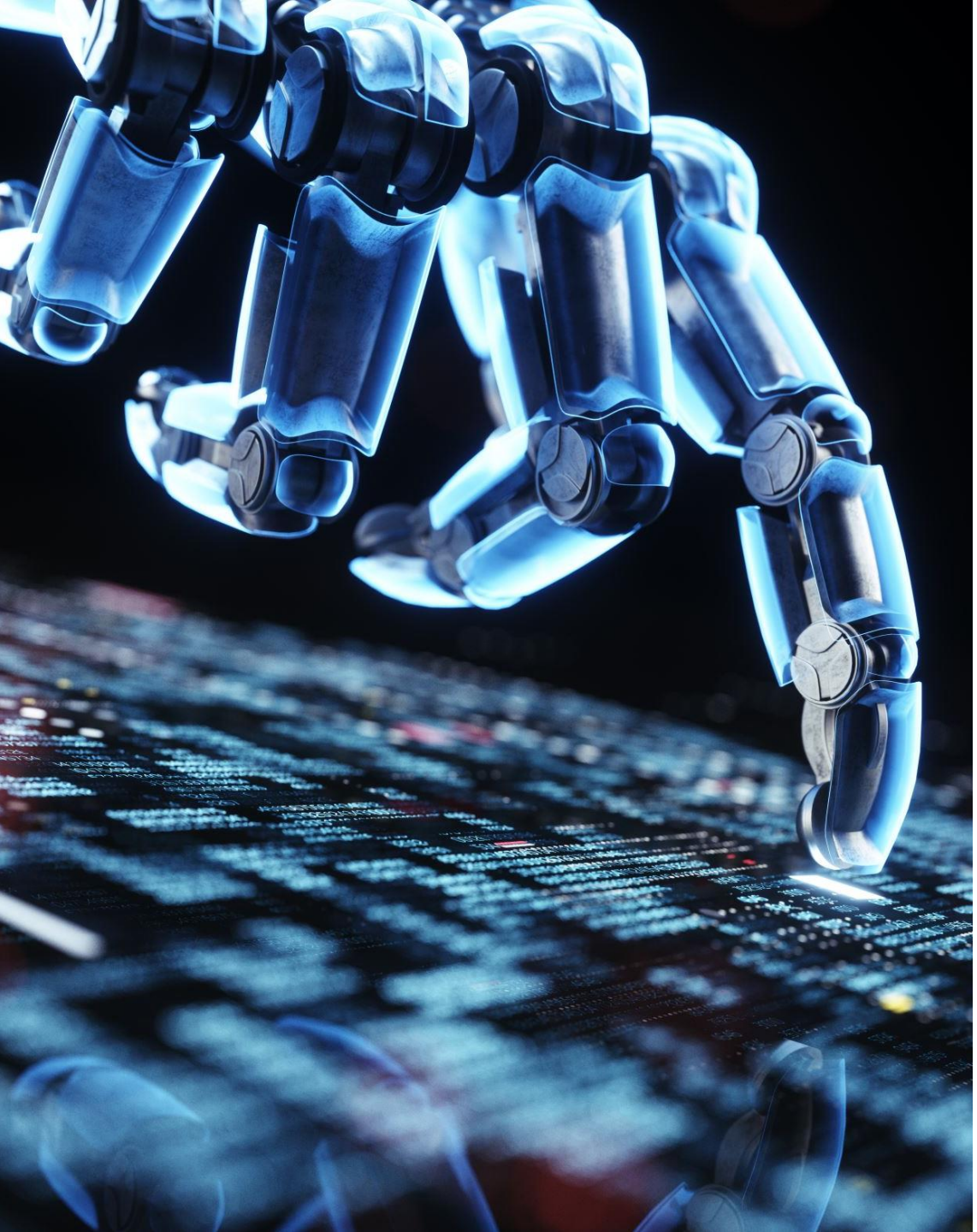
Essentialism & Technology





Essentialism & Technology





Essentialism functions as a technology that…

- Manages human difference
- Legitimates violence and exclusion while concealing its nature and function (think Buck v. Bell).
- Positions some people as human and others as ‘non-human’ or ‘other’.
- Creates and maintains injustice

Essentialism & (in)justice

- History of essentialist claims being used to maintain or create injustice
- Dominant essentialist beliefs were developed by those with power to justify their position in society and forgive/normalize injustice.

    - **Literature** – who's on the author list? who are they speaking on behalf of?
    - **Expert opinion** – who tends to be recognized at this level?
    - **Professional societies** – who belongs to these?



Essentialism & engagement

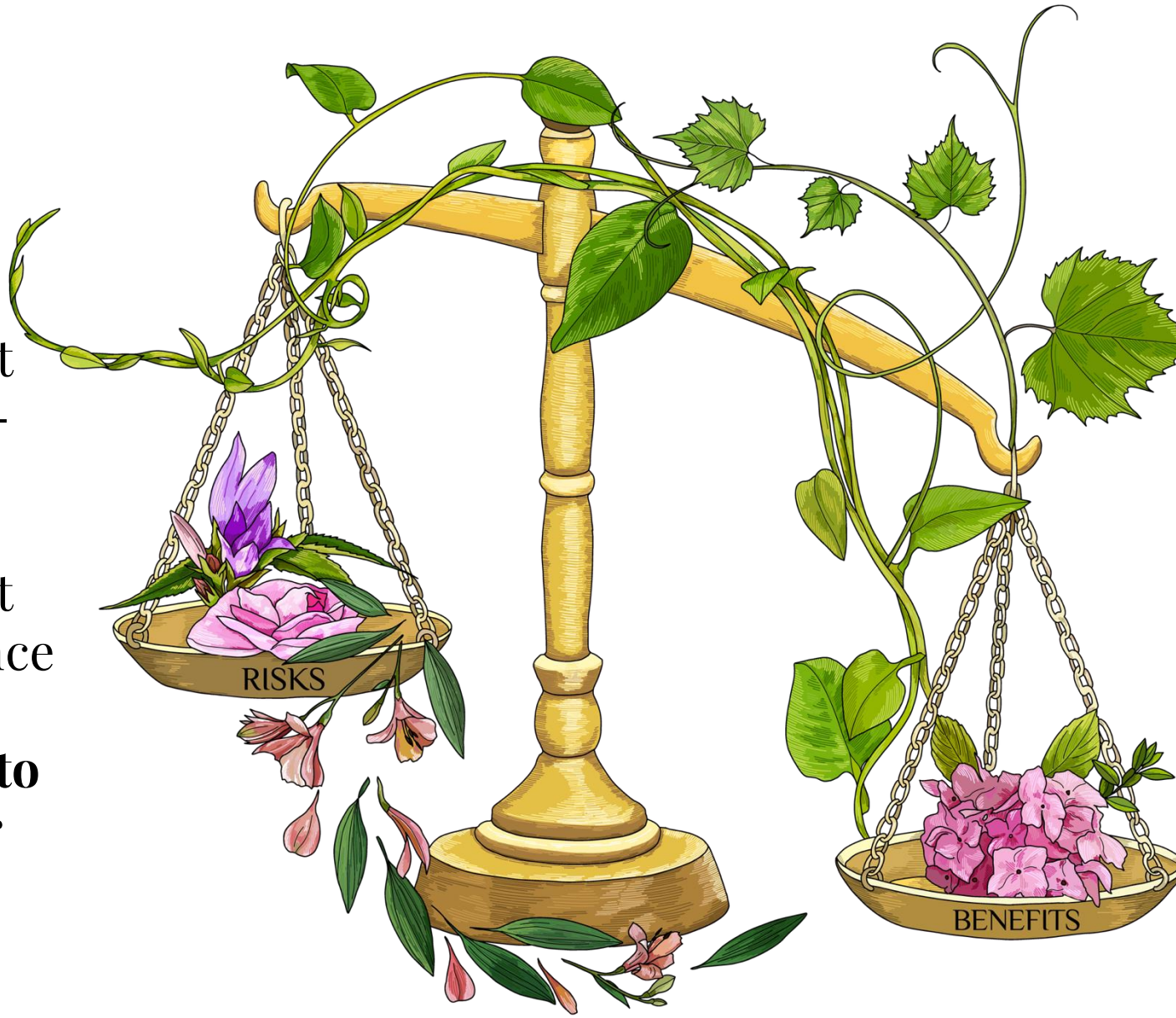


"...people thought certain groups had certain genes and had certain intellectual limits—and considering that history, dialogue between the public and researchers is really valuable...In an effort to continue examining the ethical considerations of SBG, taking into account the opinions of nonscientists is essential."

– CSB member

In summary…

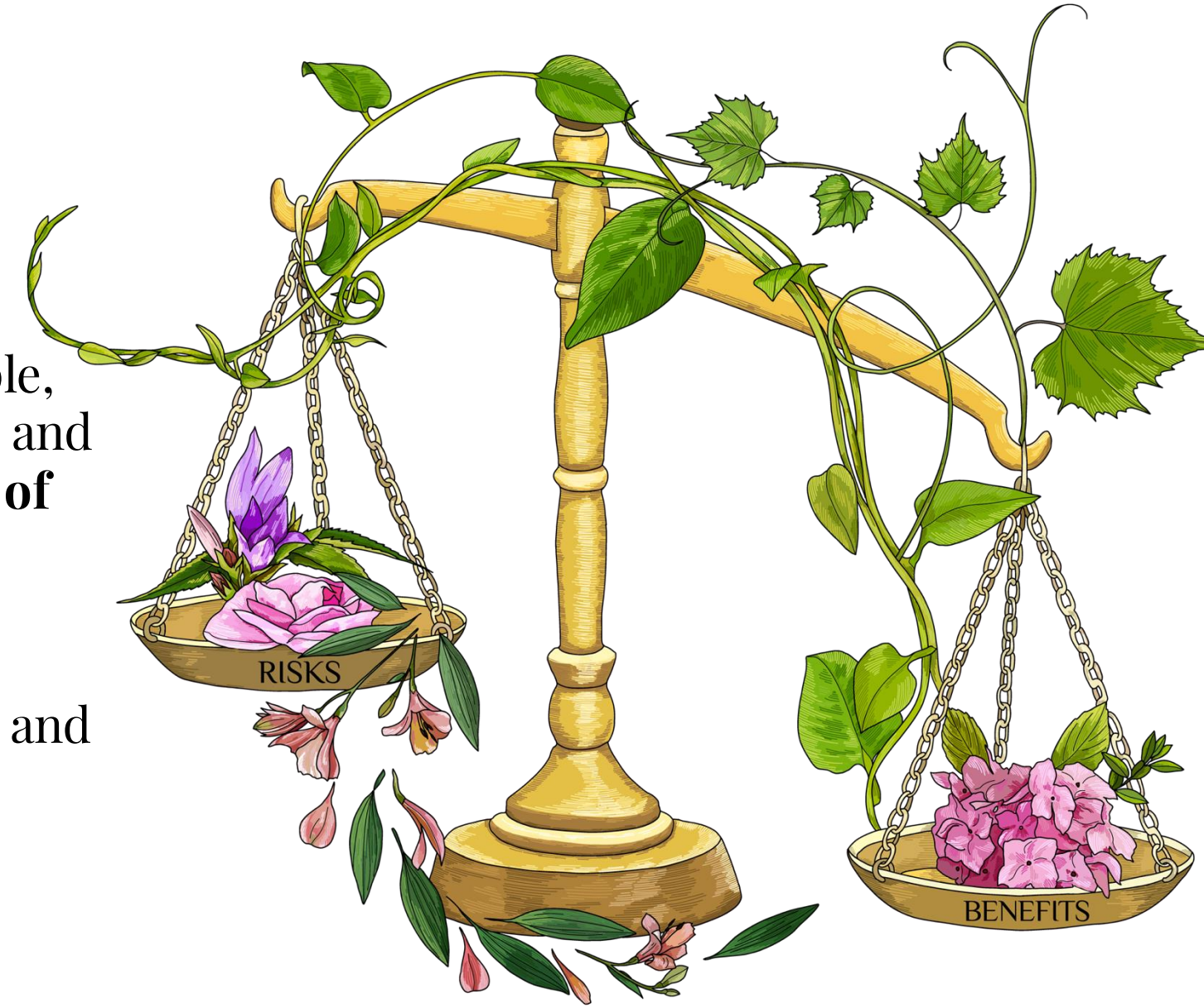
- Longstanding social narratives exist about people and their health, well-being, and capabilities.
- Longstanding social narratives exist about the origins of human difference
- **Science & technology can be used to bolster** existing social narratives **or create** new ones

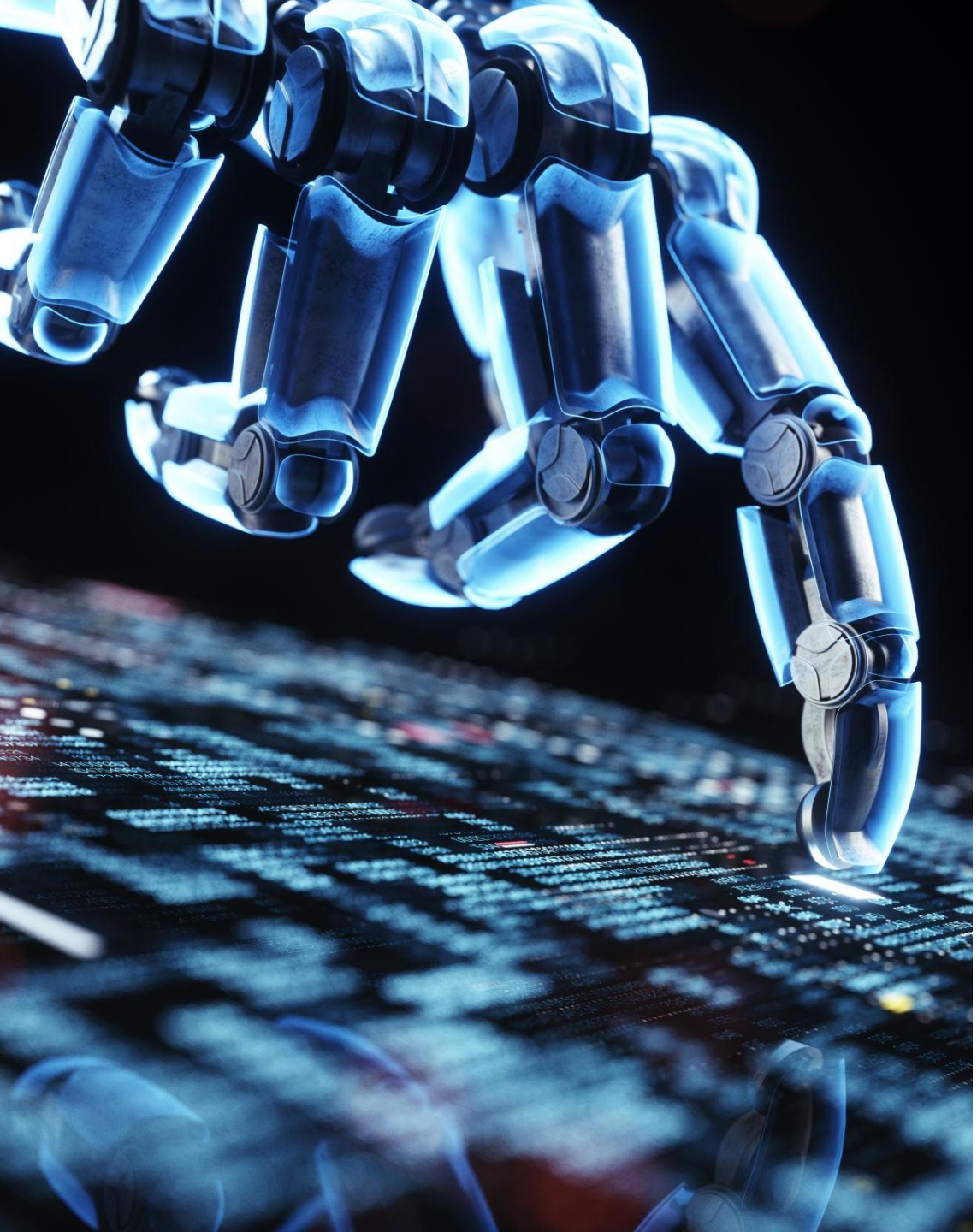


In summary…

- Essentialist narratives about people, their capabilities, their behaviors, and their health **shape the conditions of daily life**

- These narratives have **real-world repercussions** for people's health and well-being.





Looking ahead to Wednesday

What role may essentialist narratives play **as a social determinant of health**?

Share a question you'd like to discuss in section and upvote questions that you think are most important

Nobody has responded yet.

Hang tight! Responses are coming in.

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